GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Computer Engineering / Computer Science and Engineering

Subject Code: DI05000131

Subject Name: Client - Side Scripting Language (TypeScript)

W.E. F. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-04

Prerequisite:	Students should have basic knowledge of: • HTML and CSS for basic web page design • JavaScript fundamentals including DOM manipulation
Rationale:	Modern web applications require scalable, maintainable, and type-safe code. TypeScript enhances JavaScript by adding static typing, modern object-oriented features, and strong tooling support, and it is widely used in frameworks such as Angular, React, and Node.js. This course helps students transition from basic JavaScript to professional front-end development practices, enabling them to build structured and maintainable web applications used in industry.

Course Outcome:

After Completion of the Course, Student will be able to:

No	Course Outcomes	RBT Level
01	Use TypeScript data types, variables, and functions to develop programs	Apply
02	Develop programs using TypeScript functions and advanced type features	Apply
03	Implement object-oriented programming concepts using classes and interfaces in TypeScript	Apply
04	Develop dynamic frontend applications by integrating TypeScript with backend services and APIs.	Apply

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
0	2	4	4	0	0	20	30	50

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
1.	**Introduction to TypeScript:** • Need for TypeScript in modern web development • Limitations of JavaScript • Features of TypeScript • Installing Node.js and TypeScript • TypeScript Compiler (tsc) • Writing and compiling first TypeScript program • TypeScript configuration (tsconfig.json) • Using TypeScript in browser applications	5	17
2.	**TypeScript Basics** • Basic Data Types ○ number ○ string ○ boolean ○ any ○ unknown ○ void ○ null & undefined • Type inference • Variables and constants • Basic Operators (Arithmetic, Logical, Comparison) • Arrays and Tuples • Union Types • Type Aliases • Enums	5	17

3.	**Functions and Advanced Types** • Functions in TypeScript • Function parameters ▪ **optional parameters** ▪ **default parameters** ▪ **rest parameters** • Arrow functions • Function overloading • Type assertions • Introduction to Generics (conceptual understanding) • Error handling using try-catch • Compile-time vs runtime errors	6	20
4.	**Object-Oriented Programming and Code Organization** • Classes and Objects • Constructors • Access Modifiers ▪ public ▪ private ▪ protected • Getters and Setters • Inheritance • Interfaces • Abstract classes • Static members • Modules: Import and Export • Namespace vs Modules • Real-world examples (Student/Product systems)	6	20

5.	**Modules, DOM Programming, and Backend Integration** TypeScript with HTML and DOM Manipulation ● DOM manipulation using TypeScript ● Accessing and modifying HTML elements Event Handling and Form Processing ● Typed events (MouseEvent, KeyboardEvent) ● Working with HTML forms ● Form input handling and validation (basic) Fetch API and JSON Handling ● Working with API responses ● JSON parsing and handling ● Displaying dynamic data in HTML Backend and API Concepts ● Concept of backend in web applications ● Communication between frontend and backend using APIs ● Overview of client-server architecture Backend Integration and Database Connectivity ● Using any backend technology *(Recommended: PHP)* ● Database interaction with MySQL ● Retrieving data from database ● Returning JSON from backend Frontend–Backend Integration ● Fetching data using TypeScript (Fetch API) ● Handling API responses ● Displaying data dynamically in frontend	8	26
	Total	**30**	**100**

References/Suggested Learning Resources:

(a) Books:

1. Programming TypeScript: Making Your JavaScript Applications Scale, Boris Cherny, O'Reilly Media, 2019, ISBN: 978-1492037651.
2. Learning TypeScript: Enhance Your Web Development Skills Using Type-Safe JavaScript, Josh Goldberg, O'Reilly Media, 2022, ISBN: 978-1098110338.
3. TypeScript Quickly, Yakov Fain and Anton Moiseev, Manning Publications, 2020, ISBN: 978-1617295942.
4. Effective TypeScript: 62 Specific Ways to Improve Your TypeScript, Dan Vanderkam, O'Reilly Media, 2019, ISBN: 978-1492053743.
5. Beginning TypeScript: From Novice to Professional, Ben Grynhaus, Apress, 2021, ISBN: 978-1484262392.

(b) Open source software and website:

1. **TypeScript Official Documentation** – Comprehensive guide to TypeScript concepts and features.
   https://www.typescriptlang.org/docs/
2. **Microsoft Learn – TypeScript Fundamentals** – Beginner-friendly introduction to TypeScript programming.
   https://learn.microsoft.com/en-us/training/modules/typescript-get-started/
3. **W3Schools – TypeScript Tutorial** – Practical examples and exercises for beginners.
   https://www.w3schools.com/typescript/
4. **GeeksforGeeks – TypeScript Tutorial** – Covers TypeScript basics, OOP concepts, and examples.
   https://www.geeksforgeeks.org/typescript/
5. **tutorialspoint – TypeScript Tutorial** – Step-by-step explanation of TypeScript features and syntax.
   https://www.tutorialspoint.com/typescript/index.htm
6. **FreeCodeCamp – TypeScript Full Course** – Beginner-friendly course with practical examples.
   https://www.freecodecamp.org/news/learn-typescript-beginners-guide/
7. **MDN Web Docs – JavaScript and TypeScript Resources** – Documentation on JavaScript and related technologies used with TypeScript.
   https://developer.mozilla.org/
8. **TypeScript Playground** – Online editor for writing and testing TypeScript code.
   https://www.typescriptlang.org/play
9. **Node.js Official Website** – Required runtime environment for TypeScript development.
   https://nodejs.org/
10. **GitHub – TypeScript Repository** – Official TypeScript project and examples.
    https://github.com/microsoft/TypeScript

Suggested Course Practical List:

1. Install Node.js and TypeScript, create a simple TypeScript program to display a message and execute it using the TypeScript compiler.
2. Write a TypeScript program demonstrating basic data types (number, string, boolean, any) to store and display student information.
3. Write a TypeScript program to perform operations on arrays and tuples to store and display product details.
4. Write a TypeScript program to create reusable functions (including arrow functions) to perform basic arithmetic operations.
5. Write a TypeScript program to implement classes and objects to store and display student details.
6. Write a TypeScript program demonstrating interfaces and inheritance to model a simple employee management example.
7. Write a TypeScript program to create modules using export and import to perform mathematical operations.
8. Create a web form and write a TypeScript program to validate user inputs such as name, email, and age.
9. Write a TypeScript program to fetch data from a public API and display the results on a webpage.
10. Develop a simple web application (e.g., To-Do List or Task Manager) using TypeScript demonstrating classes, arrays, and DOM manipulation.

List of Laboratory/Learning Resources Required:

1. **Computer System Configuration**: Desktop or Laptop with minimum specifications:
    - Processor: Intel i3 (or equivalent) and above
    - RAM: 8 GB (recommended for database and virtualization tasks)
    - Storage: Minimum 256 GB SSD / HDD
    - Operating System: Windows 10/11, Ubuntu Linux, or macOS
2. **Software Requirements**
    - **Node.js – JavaScript Runtime Environment**
        - Download: https://nodejs.org/
        - Used to run TypeScript applications and manage packages
    - **TypeScript Compiler (tsc)**

- Install using npm: npm install -g typescript
- Used to compile TypeScript code into JavaScript

- **Visual Studio Code – Code Editor**
  - Download: https://code.visualstudio.com/
  - Recommended extensions: TypeScript, ESLint, Prettier
- **Web Browser (Google Chrome / Mozilla Firefox)**
  - Used to test web applications and DOM-based programs
- **Node Package Manager (npm)**
  - Used to manage libraries and dependencies in TypeScript projects
- **TypeScript Playground (Online Editor)**
  - https://www.typescriptlang.org/play
  - Useful for experimenting with TypeScript code without installation

Suggested Project List:

1. **Student Management System**

   Design a TypeScript-based application to manage student details, including adding, updating, and displaying student information using classes and arrays.

2. **To-Do List / Task Manager Application**

   Develop a web-based task manager using TypeScript to add, display, update, and delete tasks with DOM manipulation and event handling.

3. **Product Catalog Management System**

   Create a TypeScript application to manage product details such as product ID, name, price, and category using interfaces and object-oriented programming concepts.

4. **Simple Calculator Web Application**

   Build a calculator using TypeScript that performs arithmetic operations and demonstrates reusable functions and event handling.

5. **Form Validation System**

Develop a TypeScript-based form validation system that verifies user inputs such as name, email, password, and age before form submission.

6. **Weather Information Dashboard**

   Create a web application that fetches weather data from a public API and displays location-based weather information dynamically on a webpage.

7. **User Data Viewer using Public API**

   Fetch and display user information from an external API and present it in a structured table format using TypeScript and DOM manipulation.

8. **Expense Tracker Application**

   Develop a small TypeScript application to record daily expenses, calculate totals, and display expense reports.

9. **Simple Quiz Application**

   Build an interactive quiz system using TypeScript that presents questions, evaluates answers, and calculates scores.

10. **Online Book Catalog Viewer**

    Develop a TypeScript-based web application to display book details from a dataset or API and allow users to search or filter books.

Beyond these projects, students should:

- Practice TypeScript programming concepts such as types, interfaces, classes, and modules.
- Develop interactive web pages using TypeScript and DOM manipulation.
- Explore integration with public APIs for real-time data retrieval.
- Understand how TypeScript improves maintainability and scalability of JavaScript applications.
- Build small web applications that demonstrate modular and reusable code design.

Suggested Activities for Students:

In addition to classroom and laboratory sessions, students are encouraged to participate in the following **co-curricular and experiential learning activities**. These activities strengthen conceptual understanding, teamwork, and practical development skills using TypeScript.

Recommended Activities

a) **Hands-on Practical Sessions in a Controlled Lab Environment**

- Conduct structured programming exercises demonstrating **TypeScript syntax, types, functions, classes, and modules**.
- Practice **DOM manipulation, event handling, and form validation** using TypeScript.

b) **Web Application Development Exercises**

- Develop small web applications such as **task managers, calculators, or student management systems**.
- Implement features such as **data handling, user interaction, and modular programming**.

c) **Integration with Web APIs**

- Fetch and display real-time data from public APIs such as weather services or user datasets.
- Explore how TypeScript improves reliability and error detection in API-based applications.

d) Code Quality and Maintainability Practices

- Apply TypeScript's static typing features to reduce runtime errors.
- Organize programs using modules, interfaces, and object-oriented design for better maintainability.

e) **Study and Presentation of Modern Web Development Practices**

- Study how TypeScript is used in modern frameworks such as Angular, React, and Node.js applications.
- Present findings on benefits of TypeScript in large-scale software development and frontend engineering.